INCH-POUND

MIL-G-24748A(SH)
12 January 1994
SUPERSEDING
(See 6 5)

MILITARY SPECIFICATION

GASKETS, METALLIC AND RUBBER, FOR USE IN UNION PIPE FITTINGS 1/4 TO 2-INCH NPS WITH VARIOUS FLUIDS FROM -30 °F (-34 °C) TO 400 °F (204 °C)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gaskets, metallic and rubber, consisting of a nitrile or ethylene-propylene rubber inner ring attached to a copper-alloy cast outer metallic ring for use as a static seal in union connections of piping systems in air, water, steam, and petroleum products.

1.2 Classification. Gaskets shall be of the following types, categorized by the rubber portion of the gasket (see 3.2.2), as specified (see 6.2)

a Type A - NBR (Nitrile rubber) inner ring
b. Type B - EPM (Ethylene-propylene rubber) inner ring

1.2.1 Sizes. Gaskets shall conform to the dimensions shown in 3.3.1 for the specified nominal size required (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5330
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SPECIFICATIONS

FEDERAL

QQ-C-390	Copper Alloy Castings (Including Cast Bar)

MILITARY

MIL-F-1183	Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for
MIL-P-4861	Packing, Preformed, Rubber · Packaging of
MIL-F-24227	Fittings and Flanges, Cast Bronze, Silver Brazing, Suitable for Ultrasonic Inspection

STANDARDS

MILITARY

MIL-STD-289	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-298	Visual Inspection Guide for Rubber Extruded Goods

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 61	Standard Specification for Steam or Valve Bronze Castings, (DOD adopted)
B 62	Standard Specification for Composition Bronze or Ounce Metal Castings; (DOD adopted)
D 395	Standard Test Methods for Rubber Property - Compression Set, (DOD adopted)

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D 412	Standard Test Methods for Rubber Properties in Tension; (DOD adopted)
D 471	Standard Test Method for Rubber Property - Effect of Liquids; (DOD adopted)
D 573	Standard Test Method for Rubber - Deterioration In An Air Oven; (DOD adopted)
D 2240	Standard Test Method for Rubber Property - Durometer Hardness, (DOD adopted)
E 145	Standard Specification for Gravity-Convection and Forced-Ventilation Ovens

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3, prior to delivery of any gaskets

3.2 Materials.

3.2.1 Metallic outer ring. Material for the metallic outer rings for classification types A or B shall be in accordance with Unified Numbering System (UNS) numbers C83600. C90300, or C92200. Typical specifications for these materials include, but are not limited to, ASTM B 61, ASTM B 62, or QQ-C-390 Powdered metal processes are permitted

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3.2.2 Rubber inner ring. Rubber used for the inner ring of each gasket classification type shall conform to 3.2.2.1 and 3.2.2.2. For Type B gaskets the rubber shall be a compound using an ethylene-propylene polymer, either ethylene-propylene copolymer or ethylene-propylene-diene terpolymer, which will meet the requirements herein.

3.2.2.1 Temperature range. Each gasket type shall be suitable for continuous use in the temperature range indicated below (see 6.1).

Type A = -30 degrees Fahrenheit (°F) (-34 degrees Celsius (°C)) to 250 °F (121 °C)

Type B = -30 °F (-34 °C) to 400 °F (204 °C) (Steam) or to 300 °F (149 °C) (air)

3.2.2.2 Physical properties. The rubber portion of each gasket type shall conform to the physical properties of table I Physical properties shall be determined as specified (see 4.3.1).

TABLE I. *Physical properties.*

Property and condition	Limitations	
	Type A	Type B
Original:		
Tensile strength, min, lb/in^2 (MPa)	2031 (14)	1500 (10 3)
Ultimate elongation, min , percent	250	150
Durometer hardness (h), ±5 points	70	75
Heat resistance: 70h at 212 °F (100 °C)		
Change in tensile strength, percent	±30	±15
Change in ultimate elongation, max percent	—50	—30
Change in durometer hardness, points	±15	±10
Compression set Method B, max, percent 22h at 158 °F (70 °C)	25	25
Water resistance: 70h at 212 °F (100 °C)		
Change in hardness points	±10	—
Change in volume, percent	±15	±5

(Table I Continued on Following Page)

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TABLE I. *Physical properties - Continued.*

Property and condition	Limitations	
	Type A	Type B
Fluid resistance·		
Fuel A 70h at 73 °F (23 °C)		
Change in tensile strength, max percent	—25	—
Change in ultimate elongation, max, percent	—25	—
Change in durometer hardness, points	±10	—
Change in volume, percent	—5 to +10	—
Fuel B 70h at 73 °F (23 °C)		
Change in tensile strength, max, percent	—60	—
Change in ultimate elongation, max, percent	—60	—
Change in durometer hardness, points	0 to —30	—
Change in volume, percent	0 to +40	—
No 1 Oil 70h at 212 °F (100 °C)		
Change in tensile strength, max, percent	—25	—
Change in ultimate elongation, max, percent	—45	—
Change in durometer hardness, points	—5 to +10	—
Change in volume, percent	—10 to +5	—
No 3 Oil· 70h at 212 °F (100 °C)		
Change in tensile strength, max, percent	—45	—
Change in ultimate elongation, max, percent	—45	—
Change in durometer hardness, points	—10 to +5	—
Change in volume, percent	0 to +25	—
Steam resistance 70h at 400 °F (204 °C)		
Change in tensile strength, max, percent	—	—50
Change in ultimate elongation, max, percent	—	—40
Change in durometer hardness, points	—	±5
Change in volume, percent	—	0 to 15
Change in compression set, max, percent	—	50

3.2.3 Rubber-to-metal adhesives. Adhesives used to attach the inner rubber ring to the outer metallic ring, if any, shall be selected for compatibility with the materials of the specific gasket and shall be properly cured.

3.2.4 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed

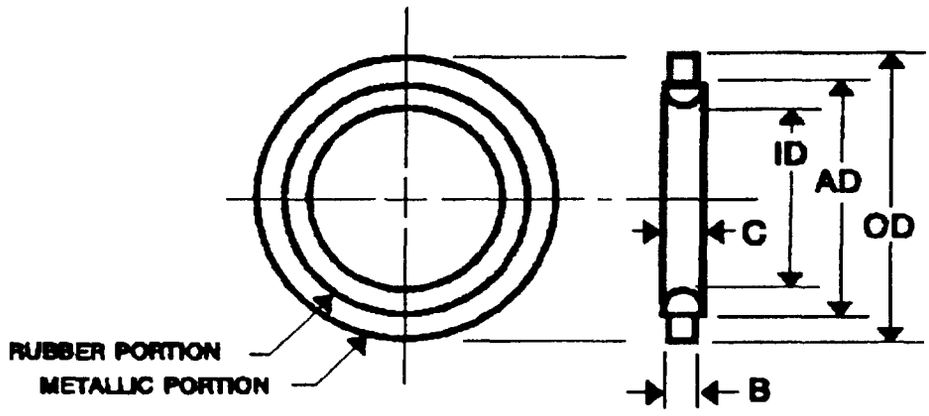
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to virgin raw materials. None of the above shall be interpreted to mean that the use of used products is allowed under this specification unless otherwise specifically specified.

3.3 Design and fabrication requirements.

3.3.1 Designs. Gaskets shall be designed for use with standard union fittings for the specified nominal pipe size and conform to the basic dimensional requirements of table II The cross-section for the rubber portion of the gasket may be circular or elliptical.

TABLE II. *Gasket dimensions.*



Nominal pipe size	Metallic Portion			Rubber portion	
	OD (+0 005)	AD (±1/64)	B (±0 005)	ID	C
0 250	1 008	13/16	0 050	0 676 ± 0 009	0 072/0 100
0 375	1 184	15/16	0 050	0 801 ± 0 009	0 072/0 100
0 500	1 394	1-3/16	0 050	1 051 ± 0 010	0 072/0 100
0 750	1 715	1-1/2	0 060	1 299 ± 0 012	0 100/0 120
1 000	2 045	1-3/4	0 070	1 549 ± 0 015	0 100/0 130
1.250	2 471	2-1/16	0 070	1 862 ± 0 015	0 100/0 130
1 500	2 791	2-3/8	0 070	2 175 ± 0 017	0 100/0 130
2 000	3 391	3	0 090	2 734 ± 0 024	0 135/0 160

NOTES.

1. Dimensions are in inches
2. Surface finish (exposed surfaces) shall be 70 roughness average (RA) or smoother
3. Break sharp edges

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3.3.2 Fabrication. The rubber inner ring shall be molded in place within or onto the metallic outer ring utilizing a covulcanization chemical bond during the molding process.

3.4 Identification. All gaskets shall be individually packaged and each package shall have the following information printed on the outside (see 5.1):

a. National stock number
b. Nomenclature (gasket, metallic and rubber: size___nps, rubber material ___ - specify NBR or EPM)
c. Specification number
d. Manufacturer's name/trademark
e. Manufacturer's rubber compound number
f. Manufacturer's batch number (see 6.6.1)
g. Contract date
h. Rubber cure date (see 6.6.2)
i. Shelf life (see 6.6.3).
j. Part or identifying number (see 6.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall include base material tests specified in 4 3 1 and final product tests specified in 4.3.2 (see 6.3 and appendix). The base material tests shall be performed once for each of the base materials used for the gasket construction to demonstrate acceptability of the base materials to the requirements of 3 2. The final product tests shall be performed for each size range, and type of gaskets supplied to demonstrate suitability of the finished product for its intended use. All tests and inspections shall be performed on sample units produced with equipment and procedures normally used in production.

4.3.1 Base material tests.

4.3.1.1 Sampling for base material tests. Sufficient test coupons, plugs, finished gaskets, and so forth, of the required shape and size shall be provided for performance of the tests specified in 4.3 1.2 and 4 3 1 3.

4.3.1.2 Metallic outer ring material. The metal shall be tested for conformance to the chemical and mechanical properties specified in the material specification used (see appendix)

4.3.1.3 Rubber inner ring material. The rubber shall be tested by each of the methods enumerated in 4.3.1.4 for conformance to the physical properties identified therein.

4.3.1.4 Physical properties. The rubber portion of each gasket type shall conform to the physical properties of table I. Physical properties shall be determined in accordance with the test methods for rubber products as follows.

Property	Test method
Tensile strength	ASTM D 412, die C
Ultimate elongation	ASTM D 412, die C
Durometer hardness	ASTM D 2240
Heat resistance	ASTM D 573
Compression set	ASTM D 395
Water resistance	ASTM D 471, distilled water
Fluid resistance (type I)	ASTM D 471
Steam resistance (type II)	4 3.1.4 1

4.3.1.4.1 Steam resistance test. Test shall be performed in accordance with ASTM D 573 with the following supplemental requirements:

a *Test oven.* Type IIA as specified in ASTM E 145

b. *Test container.* A container shall be used in the test oven with sufficient volume and provision for vertically suspending test specimens over a water reservoir

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c *Specimen arrangement.* The specimens shall be arranged such that they will not touch each other, the sides of the container nor the water.

d. *Water reservoir.* Water shall be distilled and of sufficient quantity to last the entire duration of the test.

4.3.2 Final product tests.

4.3.2.1 Test size ranges. Gaskets shall be considered to fall within one of three basic size ranges as defined by the thickness of its rubber portion (see table II). The three size ranges include the 1/4 through 1/2-inch nominal size, the 3/4 through 1-1/2-inch nominal size, and the 2-inch nominal size, inclusive. Performance of final product tests for any size of a specific gasket type shall extend to all sizes within the applicable size range of the same gasket type

4.3.2.2 Sampling for final product tests. Two gaskets within each size range and classification, manufactured in the same facility where production gaskets will be processed, shall be inspected and tested as specified in 4.3.2.3 through 4.3.2.6.

4.3.2.3 Dimensional inspection. Each sample gasket shall be dimensionally inspected for conformance to dimensions OD, B, ID, and C of table II The actual dimensions of each gasket shall be noted.

4.3.2.4 Visual inspection. Each sample gasket shall be visually inspected to ensure the surface finish of exposed surfaces of the metallic portion is 70 RA or better, and that the exposed surfaces of the rubber portion are as specified in 4 4.2.1. Each sample shall also be visually inspected to ensure there is no visible separation between the rubber and metallic portions.

4.3.2.5 Gasket assembly test. Each sample gasket shall be assembled into a standard union connection and tested as follows to demonstrate suitability for use in confined spaces.

a. *Test Apparatus.* The test apparatus shall be a standard straight union fitting per MIL-F-1183, except that the male face of the tailpiece shall be intentionally damaged to defeat the effectiveness of the union's primary metal-to-metal sealing surface (the gasket seating surface shall not be damaged) The test apparatus nominal pipe size shall be consistent with the gasket nominal pipe size to be tested, within a given test size range (see 4 3 2 1). A hydrostatic pressure leakage test similar to that of 4.3 2 5 d may be performed without a gasket installed to demonstrate that this objective has been met. The manner of achieving this objective is optional, but should not result in raised metal which could cut or otherwise damage the rubber portion of the gasket during assembly (see appendix)

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b *Installation test.* Position and restrain the tailpiece and thread-piece of the union so that they are axially aligned and there is a 0 065-inch (for 1/4 through 1/2-inch nominal size), 0 090-inch (for 3/4 through 1-1/2-inch nominal size) or 0.125-inch (for 2-inch nominal size) maximum clearance between the end of the tail-piece and the O-ring seating surface of the threadpiece Measure and record the actual clearance used Install the gasket by sliding it between the restrained halves of the union until it is centered in its normal assembled position. Remove the tailpiece and threadpiece restraint and tighten the union connection.

c. *Air leakage test.* Subject the assembled union to the air pressure leakage test and acceptance criteria as specified in MIL-F-1183.

d. *Hydrostatic pressure leakage test.* Subject the assembled union to the hydrostatic pressure leakage test and acceptance criteria as specified in MIL-F-1183 The hydrostatic test pressure shall be 600 pounds per square inch (lb/in^2)

e. *Disassembly.* Completely disassemble the union (no axial constraint required) and remove the gasket.

4.3.2.6 Final inspection. After completing the gasket assembly test, each of the sample gaskets shall be inspected for damage to the rubber portion and for separation of the rubber portion from the metallic portion of the gasket There shall be no separation of the rubber from the metal ring.

4.4 Quality conformance inspection.

4.4.1 Sampling for quality conformance.

4.4.1.1 Inspection lot. All gaskets of the same classification and size, with the rubber portion cured in the same production run and from the same batch (see 6.6.1), presented at one time shall be considered a lot for purposes of sampling and inspection.

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4.4.1.2 Sampling plan. Sampling for gasket quality conformance shall be in accordance with table III.

TABLE III. *Sampling plan.*

Inspection lot size	Sample size
2 to 8	2
9 to 15	3
16 to 25	5
26 to 50	8
51 to 90	13
91 to 150	20
151 to 280	32
281 to 500	50
501 to 1,200	80
1,201 to 3,200	125
3,201 to 10,000	200
10,001 to 35,000	315
35,001 to 150,000	500
150,001 to 500,000	800
500,001 and over	1,250

4.4.2 Examination. Each of the sample gaskets selected as specified in 4.4.1.2 shall be examined to verify compliance with all the requirements of this specification not involving tests (see 6.3). Any gasket in the sample containing one or more defects shall be rejected (see 6 9).

4.4.2.1 Rubber portion. Exposed surfaces of the inner rubber ring shall be inspected for visual defects in accordance with MIL-STD-289 or MIL-STD-298, as applicable.

4.4.2.2 Metallic portion. Exposed surfaces of the outer metallic ring shall be inspected to verify a surface finish of 70 RA or smoother This inspection may be accomplished visually using standard surface finish gauge blocks to estimate actual surface finish.

4.4.2.3 Dimensions. The outer diameter and the thickness of the metallic portion, and the inside diameter and the thickness of the rubber portion, on each sample gasket shall be measured and shall be in accordance with the requirements specified in 3.3 1

4.4.2.4 Rubber-to-metal attachment. Each sample gasket shall be visually inspected to verify attachment of rubber-to-metal is along the entire inner surface of the metallic ring and the entire outer surface of the rubber ring

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4.4.3 Rejected lots. If an inspection lot is rejected (see 6.9), the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Sampling for resubmittal of rejected lots shall be in accordance with table V, and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots. For resubmitted lots which are subsequently rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristics or providing a new lot

4.4.4 Examination for defects in packaging. An examination shall be made to determine that the packaging, packing, and markings comply with sections 3 and 5. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. The sample size and lot acceptance shall be in accordance with tables III and V. Shipping containers fully prepared for delivery shall be examined for closure defects

Examine	Defect
Packaging	Not the level specified. Not packaged as specified or required Packaging material, closures not as specified
Packing	Not the level specified; not in accordance with contract requirements
	Container not as specified, closures not accomplished by specified or required methods or materials
	Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability.
	Inadequate application of components, such as incomplete closure of case liners, containing flaps loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings, as applicable, omitted, illegible, incorrect, incomplete or not in accordance with contract requirements. Date of cure, storage instruction missing.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing, and marking. Unless otherwise specified (see 6.2), packaging, packing, and marking (see 3.4) shall be in accordance with the applicable requirements of MIL-P-4861 as specified (see 6.2).

5.1.1 Cleaning. Cleaning shall be in accordance with the manufacturer's best commercial practice.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Gaskets defined in this specification are suitable for use as a static seal in union connections in accordance with MIL-F-1183, MIL-F-24227, or similar applications. The gaskets are compatible with the fluid and corresponding temperature range (minimum/maximum) identified in table IV for each type. Where no limitation is provided for a given classification, that type is not recommended for service with the identified fluid. However, selection of any type for a given application is the responsibility of the user and should consider compatibility of all fluids which are or may be introduced into the system.

TABLE IV. *Recommended gasket application.*

Fluid	Classification (see 1 2)	
	Type A	Type B
Air/gas	−30 °F(−34 °C)/250 °F(121 °C)	−30 °F(−34 °C)/300 °F(149 °C)
Water	−30 °F(−34 °C)/250 °F(121 °C)	−30 °F(−34 °C)/400 °F(204 °C)[1]
Steam	—	400 °F(204 °C) maximum[1]
Hydraulic fluids	−30 °F(−34 °C)/250 °F(121 °C)	—

[1] Type B gaskets are suitable in steam and water continuous service up to 400 °F(204°C) and intermittent service up to 425 °F(218°C)

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TABLE V *Lot acceptance and rejection criteria.*[1]

Inspection lot size	Sample size	Reject number[2]		
		Quality conformance (see 4 4 2)	Rejected lot resubmittal (see 4 4 3)	Packaging (see 4 4 4)
2 to 8	2	1	1	1
9 to 15	3	1	1	1
16 to 25	5	1	1	1
26 to 50	8	1	1	1
51 to 90	13	1	1	2
91 to 150	20	2	1	2
151 to 280	32	2	1	3
281 to 500	50	3	2	4
501 to 1,200	80	4	3	6
1,201 to 3,200	125	6	4	8
3,201 to 10,000	200	8	6	11
10,001 to 35,000	315	11	9	15
35,001 to 150,000	500	15	13	22
150,001 to 500,000	800	22	19	22
500,001 and over	1,250	22	19	22

[1] If the inspection lot successfully passes the table III criteria, those units found defective during inspection shall be replaced with acceptable units prior to lot acceptance For rejected lots, see 4 4 3

[2] Reject number indicates the minimum number of defective units per sample size needed for rejection of the entire inspection lot Inspect the sample size until the reject number is reached

6.2 Acquisition requirements. Acquisition documents must specify the following:

a Title, number, and date of this specification
b. Part or identifying number (see 1.2 and 6.8)
c Applicable national stock number
d. Quantity
e Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2 2)
f When first article inspection is required (see 3.1)
g. When packaging is not in accordance with MIL-P-4861 (see 5 1)
h. Applicable levels of packaging and marking (see 5 1)

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.3 and appendix	DI-MISC-80653	Test Reports	—
4.4.2	DI-NDTI-80809A	Test/Inspection Reports	—

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Supersession data. This specification supersedes MIL-G-24748(SH) dated 30 November 1990 and all issues of limited distribution specifications MIL-G-SS490 and MIL-G-XX311.

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6.6 Definitions.

6.6.1 Batch. The quantity of material compounded on a mill or mixer at one time (see 3.4.f).

6.6.2 Cure date. The date that the item or material was altered industrially; that is, as to vulcanize (rubber) or to treat (synthetic elastomers) with heat or chemicals to make infusible (see 3.4.h)

6.6.3 Shelf life. The total period of time beginning with the date of manufacture/cure/assembly or inspection of test/restorative action that an item may remain in the combined wholesale (including manufacturer) and retail storage system and still maintain suitability for issue/use by the end user (see 3.4.i). (Shelf life is not to be confused with service life which is a measurement of anticipated total in-use time.)

6.7 Cross-reference. The gasket types covered by this specification revision and the corresponding types covered by the superseded specifications are interchangeable and may be binned together A cross-reference of old to new PIN's (see 6.8) appears in table VI.

TABLE VI *Gasket type comparability*

Type (see 1.2)	Nominal Size	Part or Identifying Number (PIN)	
		MIL-G-24748(SH)	MIL-G-24748A(SH)
"A" (NBR)	1/4	M24748-A04	M24748-NBR0 250
	3/8	-A06	-NBR0 375
	1/2	-A08	-NBR0 500
	3/4	-A12	-NBR0 750
	1	-A16	-NBR1 000
	1 1/4	-A20	-NBR1 250
	1 1/2	-A24	-NBR1 500
	2	-A32	-NBR2 000
"B" (EPM)	1/4	M24748-B04	M24748-EPM0 250
	3/8	-B06	-EPM0 375
	1/2	-B08	-EPM0 500
	3/4	-B12	-EPM0 750
	1	-B16	-EPM1 000
	1 1/4	-B20	-EPM1 250
	1 1/2	-B24	-EPM1 500
	2	-B32	-EPM2 000

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6.8 Part or identifying number (PIN). The PIN's to be used for gaskets acquired to this specification are created as follows:

M	24748	-	NBR	0.375
Prefix to indicate measurement system identification	Specification Number		Classification type (see 1.2)	Size (see table II)

Example: M24748-NBR0.375

6.9 Lot acceptance and rejection criteria. If the number of defective gaskets in any quality conformance inspection sample exceeds the reject number for that sample as specified in table V, the lot represented by the sample shall be rejected.

6.10 Subject term (key word) listing.

Copper-alloy cast
Inner ring
Outer ring
Static seal

6.11 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - SH
(Project 5330-N155)

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APPENDIX

TEST REPORTS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the test reports when specified in the contract or purchase order This appendix is mandatory only when data item description DI-MISC-80653 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. First article test report. The test report shall consist of the metallic outer ring base material report, the rubber inner ring base material report, and the final product test report. These reports may be provided as a single combined report or as three separate reports, at the contractor's option. The report shall include the specific results of all tests performed. The test report shall be submitted for contracting activity approval prior to delivery of the gaskets. Evidence of prior approval of the test report will be considered for satisfying this requirement provided such evidence clearly identifies the test report number, the nominal size and type tested, and that all required tests were satisfactorily completed without exception. Contracting activities should ensure that letters approving first article test reports clearly and specifically include sufficient information to allow use as evidence of prior approval.

30.1 Metallic outer ring material. The test report shall clearly identify the associated material specification, UNS alloy number, grade, class, and other identifying information.

30.2 Gasket assembly test. The manner of defeating the metal-to-metal sealing surface in the hydrostatic test (see 4 3 2 5.a) shall be described.

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-G-24748A(SH)	2 DOCUMENT DATE (YYMMDD) 940112

3 DOCUMENT TITLE
Gaskets Metallic and Rubber For use in Union Pipe Fittings ¼ to 2-inch NPS with Various Fluids from -30°F to 400°F

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed*)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Commander Naval Sea Systems Command	b TELEPHONE (*Include Area Code*) (1) Commercial (703)602-8595 (2) AUTOVON 332-8595
c ADDRESS (*Include Zip Code*) Code 03Q42 2531 Jefferson Davis Hwy Arlington VA 22242 5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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